Same Responses, Opposite Conclusions: How Detection Rules and Response Length Shape Measured Advice-Style Language in OLMo-2

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Abstract

Studies of how training changes language-model behavior often count marker phrases in free-form answers. We show that such counts can reverse under two choices that are rarely reported: the matching rule and the amount of text searched. Our case study is advice-style language in OLMo-2, whose post-training data and intermediate checkpoints are public. We combine 140 prompts in five conditions, 1B and 7B checkpoints from base to final model, a 120-sentence audit against LLM-consensus labels, and about 378,000 preference pairs per model. On the same 700 final-7B responses, the gap between personal-advice prompts and an emotional control for an empathy marker is $+32.3$ percentage points with exact matching, -14.7 once embedding matches are added, and $+25.3$ when only the first 64 tokens are searched. Swapping the matching rule alone moves this gap by 47 points (simultaneous 95% interval 28 to 66). In the audit, every false positive comes from the embedding rule. Across checkpoints, DPO widens the advice-specific gap for validation phrases and narrows it for structuring phrases at both model sizes. In the DPO preference data, referral and structuring phrases favor the chosen response, and modeling response length roughly halves that association without removing it. Marker comparisons should therefore report the matching rule, the text window, and length next to the headline rate. The audit labels are model-generated and all analyses are observational, so we make no claim about semantic empathy or causal training effects.

1 Introduction

Suppose a model answers "I've been feeling anxious and cannot sleep. What should I do?" with "That sounds really difficult. Here are a few steps you can try." A phrase detector counts an empathy marker and a structuring marker. Now consider the constructed fragment "A checklist may help. That sounds really useful for packing." It triggers the same empathy rule, because *that sounds really* appears in the second sentence, although the phrase appraises a checklist and not a person's feelings.

Such cases matter beyond one imperfect detector. Comparisons across checkpoints and prompt conditions often reduce free-form answers to occurrence rates. A longer answer offers more phrases and sentences to search, and an embedding rule adds matches that an exact rule would reject. A change in an occurrence rate can therefore reflect the model's response, the amount of text observed, the matching rule, or all three. Unless these are separated, an apparent behavioral trajectory may be a measurement trajectory.

This paper asks one question: *how do detection rules and response length change conclusions about advice-style language across OLMo-2 post-training?* We split it into three parts. **RQ1** asks how the detector's components and the text window change measured contrasts. **RQ2** asks how those contrasts vary with the control condition and the checkpoint. **RQ3** asks how phrase–preference associations in the DPO training data change once response length is modeled.

OLMo-2 suits this study because its final Instruct model is the product of four public data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). Both the intermediate checkpoints and the post-training data can be inspected.

We use three terms throughout. A **marker** is a specified textual feature, such as a phrase in our lexicon. A **detector** is the algorithm that decides whether a marker category occurs in a response. A detector output becomes a **proxy** when it stands in for a broader construct such as empathy. Our four categories keep the names of their lexicon

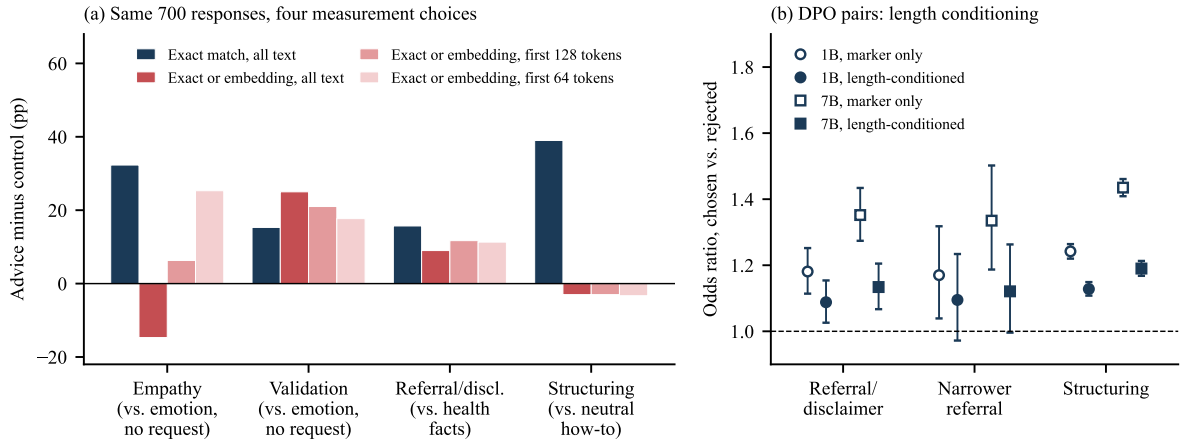


Figure 1: Two measurement choices that change the conclusion. (a) Gap between personal-advice prompts and the named control for each marker, computed on the same 700 final-7B responses under four matching rules and text windows (Table 2). The empathy gap is positive, negative, or near zero depending on the choice. (b) Odds ratio that a phrase appears in the chosen and not the rejected response of a DPO pair, before and after modeling response length (Table 7); bars are 95% intervals.

entries (*empathy-opener*, *validation-phrase*, *referral/disclaimer*, *structuring-marker*), but the names label rules and not semantic judgments.

Findings and contributions.

1. **The matching rule and the text window can flip a contrast (RQ1).** On identical final-7B responses, replacing exact matching with the exact-or-embedding union moves the empathy gap by 47.0 percentage points and the structuring gap by 42.0 (Figure 1a). In a 120-sentence audit against LLM-consensus labels, exact matching produces no false positives and every false positive of the union comes from the embedding rule.

2. **Controls and checkpoints separate markers that look alike (RQ2).** All four markers separate personal advice from general facts, yet only validation phrases exceed every targeted control. Between the SFT and DPO checkpoints, which share one prompt interface, the advice-minus-facts gap for validation widens while the gap for structuring narrows, at both sizes.

3. **Length explains about half of the preference-data association (RQ3).** In roughly 378,000 DPO pairs per model, referral/disclaimer and structuring phrases favor the chosen response. A paired model with flexible length terms shrinks these odds ratios toward one without removing them (Figure 1b); a narrower referral set is no longer distinguishable from no association.

We release prompts, the lexicon, detector outputs, audit labels, and analysis scripts.

2 Related Work

Behavior across training stages. Public checkpoint suites made training dynamics a research object (Biderman et al., 2023), and open post-training recipes such as Tulu 3 (Lambert et al., 2024) extended this to the stages after pretraining. Several recent studies compare checkpoints along such a chain. Rivière et al. (2026) trace situation modeling across OLMo-2 and Pythia stages while stress-testing construct validity. Yao et al. (2026) measure sycophancy at each stage of 13 open checkpoints by how far an injected claim moves an answer, and Huang and Cai (2026) follow 741 checkpoints from 30 lineages with planted verdicts and internal probes. Murray et al. (2026) trace unintended behaviors back to specific post-training data. Most of these studies score behavior on constrained tasks with a well-defined answer. We study the less constrained case of style markers in free-form answers, where the scoring rule itself is a degree of freedom, and we ask how much of a stagewise conclusion depends on it.

Measuring empathic and supportive language. Work on supportive language usually grounds its categories in theory and human annotation. Sharma et al. (2020) annotate 10k response pairs under

an empathy framework and train a classifier on them, and Liu et al. (2021) define support strategies for annotated conversations. Gueorguieva et al. (2026) code ten empathic tactics and find that LLM responses follow one discourse template far more often than human responses do, which is a reason to expect marker phrases to co-occur. Reviews warn that incompatible operational definitions of empathy undermine comparison (Lahnala et al., 2022). Gains on empathy-related metrics need not improve human preference (Lee et al., 2026), and warmth-oriented training is studied for its effects on accuracy and sycophancy (Ibrahim et al., 2026), outcomes that marker counts do not capture. Our detector is far simpler than these instruments, and deliberately so: lexicon and embedding rules are what a quick checkpoint comparison reaches for.

Measurement validity and LLM annotators. Behavioral testing argues for targeted controls beyond aggregate scores (Ribeiro et al., 2020). Measurement-theory accounts of evaluation separate the concept, its systematization, and the instrument that scores it (Wallach et al., 2025; Bean et al., 2025); our marker, detector, and proxy terms follow that separation. LLM judges can approach expert agreement on empathic communication in specific settings (Kumar et al., 2026), are less reliable for affective properties in mental-health text (Badawi et al., 2026), and can be compared directly with human judges (Welivita et al., 2026). Calderon et al. (2025) give a statistical procedure for deciding when an LLM may replace human annotators, which requires a human-annotated subset. Our audit has no such subset, so we use it to diagnose the detector's rules and not to validate a construct.

Length in preference learning and evaluation. Reward models and preference data encode length correlations (Singhal et al., 2023), DPO can exploit and amplify them (Park et al., 2024; Liu et al., 2024b), and length-controlled evaluation changes model rankings (Dubois et al., 2024). These papers treat length as a property of the whole response. We ask a narrower question: how much of the chosen-side advantage of specific marker phrases in the OLMo-2 DPO mixtures remains once length is modeled.

Data tracing. Exact-substring search (Liu et al., 2024a) and output tracing (Liu et al., 2025) motivate our secondary corpus analysis. Gradient-based selection (Xia et al., 2024) and probe-based attribution on OLMo-2 DPO (Xiao and Aranguri, 2026) target causal questions; our evidence is observational and does not establish which examples caused an output.

3 Measurement and Study Design

The study has five analyses, each with its own unit and scope (Table 1).

3.1 Marker categories and detector

A hand-seeded, inspectable lexicon specifies four marker categories: *empathy-opener*, *validation-phrase*, *referral/disclaimer* (artifact key `disclaimer`), and *structuring-marker* (`structure`). Two names are broader than they sound. The empathy-opener rule searches every sentence, not only an opening span. Referral/disclaimer combines referrals, disclaimers, and professional-language fragments such as "consult a" and "medical professional"; it is not a detector of referrals alone. A fifth category, `crisis_referral`, is excluded after producing 0 true positives and 11 false positives in the audit.

The detector has two components. The *exact* component is case-insensitive substring matching against the lexicon. The *embedding* component marks a response positive when any sentence reaches cosine similarity τ to a category exemplar under e5-small (Wang et al., 2024). The *union* detector fires when either component does. The primary threshold $\tau = 0.82$ was fixed before the 140-prompt battery was run and was not tuned afterwards; we also report $\tau = 0.80$ and 0.84. Appendix B gives the full specification.

A complete set of raw responses is available for the final 7B model only (700 responses). For the other checkpoints we rely on stored union-detector outputs, which cannot be recomputed under alternative rules. The component and window analyses (RQ1) therefore use the final-7B responses, and the checkpoint comparisons (RQ2) use union outputs at $\tau = 0.82$.

3.2 Prompt conditions and generation

Five conditions separate request type, emotional content, and domain: *personal advice* (60 personal or emotional help requests), *general facts* (30 knowledge questions), *emotion without a request* (20 prompts on emotional topics that do not ask for help), *neutral how-to* (20 practical requests without

Analysis	Material	Comparison	Main scope limit
RQ1: Detector and window	700 stored final-7B responses; 140 prompts, five generations	Exact-only versus union contrasts; all text versus 64- and 128-token prefixes	Generations capped at 256 tokens; final 7B only
RQ1: Sentence audit	120 sentences from 97 final responses	Detector versus LLM-consensus sentence labels	No human labels; the referral label is narrower than the detector category
RQ2: Final conditions	140 prompts, five conditions, five generations	Personal advice versus four controls at final 1B and 7B	Author-written controls are not topic-paired counterfactuals
RQ2: Checkpoint grid	90 personal-advice and general-facts prompts, five generations	Change in the advice-minus-facts gap across public 1B and 7B stages	Base-to-SFT also changes prompt serialization; detector outputs only
RQ3: DPO pairs	378,301 1B and 378,341 7B pairs	Exact chosen-versus-rejected matches, with paired flexible length terms	Observational; conditioning changes the estimand; mixtures share 21,508 pairs

Table 1: Evidence map. Each analysis answers a different part of the measurement question and has a different unit and scope limit.

emotional stakes), and *health facts* (10 impersonal clinical-information questions). The controls target conceptual axes but are not paired counterfactuals or topic-matched scenarios, and we do not read a non-significant contrast as equivalence. Each prompt receives five sampled generations (temperature 0.7, top-p 0.9, 256 new tokens, distinct seeds).

Twelve of the 140 prompts (eight personal advice, four general facts) were also used in a pilot while the lexicon was seeded, so we do not call the battery held out. Excluding them changes the final-7B advice-minus-facts gaps by at most .045 (Appendix D).

3.3 Checkpoints

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct and 7B Base→SFT→DPO→Instruct. The checkpoint grid uses the 90 personal-advice and general-facts prompts with five generations. All five conditions run on the final models and, with four generations per prompt (seeds 0–3), at Base, SFT, and DPO.

Base checkpoints have no chat template and use a plain Question/Answer format, whereas post-trained checkpoints use their official chat template. Base→SFT is therefore a stage-plus-interface transition and cannot isolate SFT training. We report it but build no claim on it; the SFT→DPO comparison shares one interface and carries the checkpoint results.

3.4 Sentence-level audit

To relate detector outputs to sentence meaning, we sample 120 final-model sentences at random within condition (40 personal advice, 30 general facts, 20 emotion without a request, 20 neutral how-to, 10 health facts), without stratifying on detector predictions. Two separately run, blinded passes of an LLM annotator (Codex) label each sentence for the four categories, and Claude Opus 4.8 adjudicates their disagreements. The artifact records the Opus identifier and run metadata; exact Codex build identifiers were not retained. These labels are a diagnostic of the detector's rules against one explicit labeling scheme. They are not human validation, and we do not use them to recalibrate response-level rates.

3.5 Inference

The clustering unit is the prompt. Each prompt contributes its mean detector-positive indicator over its generations, and condition rates are means over prompts; generations are never treated as independent trials. Between-condition and stage-by-condition tests are two-sided permutation tests on prompt-level means (10,000 permutations; Monte Carlo $p = (b+1)/(B+1)$). Benjamini–Hochberg correction is applied within each family: 24 final-condition contrasts, 28 adjacent-stage interactions on the advice-minus-facts gap, 24 stage interactions with targeted controls, and 8 DPO pairwise tests. We write p for unadjusted and q for adjusted values.

For the detector-component comparison we estimate the change in each advice-minus-control gap when exact-only matching replaces the union on the same text. We resample prompts within condition 20,000 times, keeping all five generations and both detector outputs together. Shared draws give simultaneous 95% intervals across the four changes from the maximum absolute standardized

bootstrap deviation. This analysis was added after the condition and checkpoint results were known and is exploratory.

3.6 Paired DPO length model

For each of three exact phrase groups and each model size we fit a paired choice model with conditional likelihood $\sum_i \log \sigma[\boldsymbol{\beta}^\top(\mathbf{x}_{ic} - \mathbf{x}_{ir})]$ and no intercept, where c and r index the chosen and rejected response of pair i. The features are the chosen-minus-rejected marker flag and differences in a restricted cubic spline of $\log(1 + \text{words})$ with five knots fixed at pooled percentiles before fitting (7, 78, 204, 374, and 739 words). Intervals use a prompt-cluster sandwich covariance with finite-cluster correction. The resulting conditional odds ratio (OR) equals one when the model estimates no marker–choice association at the modeled lengths. Without the length terms, the model reduces, up to a continuity correction, to the matched odds ratio of chosen-only to rejected-only pairs, which we report as the marker-only estimate.

Two cautions apply. Conditioning on an observed response property changes the estimand and does not identify a causal effect. The model specification was written down before the six fits but after the aggregate differences were known, so we treat the fits as exploratory.

Corpus searches. Searches of the pretraining, SFT, and RLVR data give secondary evidence of phrase co-occurrence (Appendix G).

Reproducibility. The artifact contains the prompts, lexicon, seeds, thresholds, detector outputs, both audit passes with adjudication reasons, paired DPO features, and the scripts that derive every reported statistic, including adjusted values. Appendix J lists the file for each analysis.

4 Results

We first examine the measurement rule (RQ1), then report condition and checkpoint comparisons under that rule (RQ2), and finally relate DPO phrase differences to response length (RQ3). The order matters: RQ1 bounds what the detector-positive rates in RQ2 can mean.

4.1 RQ1: Detector and text-window sensitivity

The matching rule changes two of four contrasts. Table 2 recomputes each targeted contrast on the same 700 final-7B responses. Under exact matching, the empathy marker is 32.3 points more frequent in personal advice than in emotional prompts without a request; under the union it is 14.7 points less frequent. The structuring marker is 39.0 points more frequent in personal advice than in neutral how-to answers under exact matching and indistinguishable under the union. The paired bootstrap tests these changes directly. Replacing the union with exact matching moves the empathy gap by $+47.0$ points (simultaneous 95% interval $[28.1, 65.9]$) and the structuring gap by $+42.0$ points ($[21.9, 62.1]$). The validation change is -9.7 points ($[-22.7, 3.3]$) and the referral/disclaimer change is $+6.7$ points ($[-14.7, 28.0]$); both intervals include zero.

The text window changes the empathy contrast again. Advice responses average 244 tokens and factual responses 187, while the three targeted controls average 251 to 255 and almost all reach the 256-token cap (Appendix D). Under the union, the empathy rates on all text are .613 for advice and .760 for emotion without a request (gap -14.7 points, $p = .088$). In the first 64 tokens they are .383 and .130 ($+25.3$ points, $p = .003$), and in the first 128 tokens .493 and .430 ($+6.3$ points, $p = .488$). Emotional-prompt responses thus acquire most of their empathy matches late in the response, consistent with the embedding rule having more sentences to test. The validation and referral/disclaimer contrasts stay positive in every window and structuring stays near zero. The window p-values are unadjusted, and only the 64-token contrast is clearly different from zero.

A stored health-facts response shows the mechanism for exact rules as well: "Here are some key points based on research findings" occurs after the first 64 tokens, so the structuring rule fires on the full text and not on the prefix.

False positives come from the embedding rule. The audit covers 120 sentences (56 from the 1B and 64 from the 7B final model, from 97 distinct responses). The two annotation passes agree on 117 sentences (97.5%); Cohen's κ is .885 for empathy expression, .663 for validation, 1.000 for referral, and .978 for structuring. Against the adjudicated labels, exact matching has no false positives. Its true-positive/false-positive/false-negative counts are 2/0/2 for empathy expression, 1/0/1 for validation, 1/0/0 for referral, and 9/0/21 for structuring. Adding the embedding rule raises structuring

Marker	Control subtracted from advice	Exact, all text	Union, all text	Union, 64	Union, 128
empathy	Emotion, no request	.323	−.147	.253	.063
validation	Emotion, no request	.153	.250	.177	.210
referral/disclaimer	Health facts	.157	.090	.113	.117
structuring	Neutral how-to	.390	−.030	−.033	−.030

Table 2: RQ1: detector and text-window sensitivity on the 700 final-7B responses. Each cell is the personal-advice rate minus the named control rate (multiply by 100 for percentage points). "All text" is the complete stored generation, capped at 256 new tokens; 64 and 128 are token prefixes. Union combines exact and embedding matches at $\tau = .82$.

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
ref./discl.	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4

Table 3: Union-detector precision, recall, and F1 against the adjudicated LLM-consensus labels for 120 sentences ($\tau = 0.82$). The referral label is narrower than the referral/disclaimer category. Response-cluster bootstrap intervals are in Appendix I.

recall from 0.30 to 0.87 and introduces every false positive in the sample (Table 3).

Two properties of the audit limit how far these numbers reach. The referral label requires an actual recommendation to seek a qualified professional, whereas the detector category also matches disclaimers and mentions of professionals, so its 0.06 precision reflects target mismatch as well as detector error. Positive labels are rare for three categories, which makes the intervals wide (Appendix I). Because errors vary by condition, subtracting two condition rates does not guarantee that they cancel.

4.2 RQ2: Controls and checkpoint comparisons

The remaining results use the union detector at $\tau = .82$, the only version available at every checkpoint. Read in light of RQ1, they describe detector outputs.

Only validation phrases exceed every control. Table 4 gives prompt-level rates per condition for the final models and Table 5 tests personal advice against each targeted control. Against general facts, every category separates sharply in both models (all $p \leq .0002$); for referral/disclaimer the rates are .900 versus .080 for 1B and .913 versus .060 for 7B. The targeted controls give a different picture for three of the four markers:

- *Empathy-opener* fires more often on emotional prompts without a request than on personal advice: .790 versus .590 for 1B ($p = .010$) and .760 versus .617 for 7B ($p = .098$).
- *Structuring* is at ceiling on neutral how-to prompts. The advice-minus-how-to difference is .000 for 1B and −.030 for 7B ($p = .574$).
- *Referral/disclaimer* fires on .740 (1B) and .720 (7B) of neutral how-to prompts and on .820 of health-facts prompts in both models. Its advice-minus-health-facts difference is .080 for 1B ($p = .352$) and .093 for 7B ($p = .159$).
- *Validation-phrase* is higher on personal advice than on every targeted control in both models (all $p \leq .0044$, all $q \leq .0082$).

The 1B and 7B profiles are close, and their small, mixed differences support no scaling claim.

Markers co-occur within responses. Among the 300 final-7B advice responses, 4.7% trigger one category, 29.0% two, 40.7% three, and 25.7% all four; none triggers zero. Referral/disclaimer is present in 93.0% of validation-positive advice responses. Pairwise phi correlations are weak (−.14 to .21), so they neither establish nor rule out a shared response template of the kind reported by Gueorguieva et al. (2026).

DPO moves the validation and structuring gaps in opposite directions. Advice-side rates that rise across stages are ambiguous, because post-training could make all outputs more cautious and list-like. We therefore track the personal-advice-minus-general-facts gap at each stage (Appendix E) and test its change between adjacent stages, a stage-by-condition difference-in-differences on per-prompt trajectories (Table 6; 60 personal-advice and 30 general-facts prompts).

Between SFT and DPO, which share the chat interface, the validation gap widens by +.213 for 1B

Marker output	Model	Personal advice	General facts	Emotion, no request	Neutral how-to	Health facts
empathy opener	1B	0.590	0.040	**0.790**	0.020	0.340
	7B	0.617	0.027	**0.760**	0.070	0.360
validation phr.	1B	0.363	0.007	0.120	0.000	0.040
	7B	0.383	0.013	0.130	0.030	0.060
ref./discl.	1B	0.900	0.080	0.510	0.740	0.820
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.580	0.820	**1.000**	0.920
	7B	0.970	0.547	0.830	**1.000**	0.840

Table 4: Mean prompt-level union-detector rate per condition (final models, $\tau = 0.82$, five generations per prompt). Bold marks the cells that contradict an advice-specific reading: the empathy-opener output peaks when emotion appears without a request, and structuring is at ceiling on neutral how-to prompts.

Marker	Model	Emotion, no request	Neutral how-to	Health facts
empathy	1B	$-.200$ (.010)	.570 (.0002)	.250 (.016)
	7B	$-.143$ (.098)	.547 (.0002)	.257 (.031)
validation	1B	.243 (.003)	.363 (.0002)	.323 (.003)
	7B	.253 (.003)	.353 (.0003)	.323 (.004)
referral	1B	.390 (.0002)	.160 (.006)	.080 (.352)
	7B	.373 (.0002)	.193 (.001)	.093 (.159)
structuring	1B	.180 (.0002)	.000 (1.0)	.080 (.018)
	7B	.140 (.004)	$-.030$ (.574)	.130 (.026)

Table 5: Personal advice minus each targeted control, with the prompt-clustered permutation p in parentheses (10k draws). After Benjamini–Hochberg correction across the 24 tests, all validation contrasts keep $q \leq .0082$. A non-significant cell means no detected difference, not equivalence.

Model	Marker	Base→SFT + interface	SFT→DPO
1B	empathy opener	$+0.190$ (.004)	$+0.117$ (.121)
1B	validation phrase	$+0.093$ (.137)	$+0.213$ (.002)
1B	referral/disclaimer	$+0.253$ (.002)	$+0.190$ (.002)
1B	structuring	$+0.337$ (.002)	-0.367 (.002)
7B	empathy opener	$+0.263$ (.002)	$+0.070$ (.387)
7B	validation phrase	$+0.103$ (.121)	$+0.147$ (.021)
7B	referral/disclaimer	$+0.243$ (.002)	$+0.083$ (.203)
7B	structuring	$+0.227$ (.002)	-0.277 (.002)

Table 6: Change in the personal-advice-minus-general-facts gap across transitions, with q in parentheses (prompt-clustered permutation tests, 10k draws, Benjamini–Hochberg across all 28 adjacent-stage tests). The Base→SFT column also changes the input serialization. Later transitions have unadjusted $p \geq .168$.

($q = .002$) and $+.147$ for 7B ($q = .021$). The structuring gap narrows by $-.367$ for 1B and $-.277$ for 7B (both $q = .002$): DPO raises structuring on factual answers while advice-side rates are already near ceiling. The 1B referral/disclaimer gap widens by $+.190$ ($q = .002$); the 7B change of $+.083$ is not significant ($q = .203$), and neither empathy-opener change is. No RLVR or final-merge gap change is significant (all unadjusted $p \geq .168$).

All eight Base-to-SFT changes are positive and six survive correction, but this transition also switches from plain question–answer text to the chat template, so it cannot be attributed to SFT. Every category is already detector-positive at Base.

Against an emotional control, DPO reverses the empathy gap. The Base/SFT/DPO runs with all five conditions test 24 planned stage-by-control interactions (Appendix F), of which four survive correction at $\tau = .82$. The clearest concerns empathy relative to emotion without a request. The 7B gap increases at the Base→SFT/interface transition ($+0.358$, $q = .016$), and DPO then reverses it at both 1B (-0.467, $q = .001$) and 7B (-0.642, $q = .001$). At 7B, SFT also shifts structuring toward neutral how-to (-0.208, $q = .026$). No validation or referral interaction reaches $q < .05$. Three interactions are significant at $\tau = .80$, four at .82, and five at .84; the two 7B empathy transitions are significant at all three. The widening advice-minus-facts gaps in Table 6 therefore do not all indicate advice-specific change: after DPO, the empathy marker tracks emotional content more than the request for advice.

4.3 RQ3: Phrase preference and response length

The DPO mixtures contain 378,301 preference pairs for 1B and 378,341 for 7B. In both, referral/disclaimer and structuring phrases appear more often in the chosen response than in the rejected one, with matched odds ratios from 1.17 to 1.44 across the three phrase groups (Table 7, marker only).

The raw difference follows the longer response. Only 19.1% (1B) and 18.7% (7B) of pairs have responses within ± 20 words of each other. Where

Mix	Phrase group	Marker only	With length terms
1B	referral/discl.	1.18 [1.11, 1.25]	1.09 [1.03, 1.15]
1B	narrower referral	1.17 [1.04, 1.32]	1.10 [0.97, 1.23]
1B	structuring	1.24 [1.22, 1.26]	1.13 [1.11, 1.15]
7B	referral/discl.	1.35 [1.27, 1.43]	1.13 [1.07, 1.21]
7B	narrower referral	1.34 [1.19, 1.50]	1.12 [1.00, 1.26]
7B	structuring	1.44 [1.41, 1.46]	1.19 [1.17, 1.21]

Table 7: RQ3: odds ratio [95% interval] that a phrase group appears in the chosen and not the rejected response of a DPO pair. "Marker only" is the matched odds ratio; "with length terms" is the paired conditional model with flexible length terms and prompt-cluster intervals. The 7B narrower-referral lower bound is 0.996. Three-decimal values are in Appendix H.

the chosen response is longer by more than 20 words, the structuring odds ratio is 2.36 (1B) and 2.61 (7B); where the rejected response is longer, it is 0.58 and 0.62. Within ± 20 words the broad-referral difference is near zero and the structuring difference is positive but small (Appendix H).

Length terms remove about half of the association. In the paired model with flexible length terms, the broad referral/disclaimer odds ratio falls from 1.18 to 1.09 for 1B and from 1.35 to 1.13 for 7B, and structuring falls from 1.24 to 1.13 and from 1.44 to 1.19. All four intervals still exclude one. For the narrower referral set, which requires an explicit doctor or professional object, both intervals include one. An odds ratio of one means no modeled marker–choice association; it is not a statement about a change in choice probability.

One-pair-per-prompt and leave-largest-source-out checks preserve the broad-referral and structuring directions. The two mixtures share 21,508 content-identical pairs, so the 1B and 7B results are not independent replications. Source heterogeneity and the forest plot are in Appendix H.

5 Discussion

What changes with the measurement, and what does not. Two conclusions depend on the measurement choice. The empathy contrast against emotional prompts is positive under exact matching, negative under the union, and positive again in a 64-token window. The structuring contrast against neutral how-to prompts is large under exact matching and absent under the union. Three conclusions do not move in our checks: validation phrases exceed their strongest targeted control at all three thresholds, DPO reverses the 7B empathy gap against emotional prompts at all three thresholds, and no RLVR or final-merge gap change is significant. Stability under three thresholds is a property of the detector and does not establish construct validity.

Why the union detector behaves this way. The embedding rule fires when any sentence resembles an exemplar, so its chance of firing grows with the number of sentences. Final-7B responses to the targeted controls are as long as advice responses and close to the 256-token cap, while factual responses are shorter. A rule with this property separates long from short conditions well and separates two long conditions poorly, which matches the pattern in Tables 2 and 5. Exact matching has the opposite profile in the audit: no false positives and low recall.

Recommendations. For marker-based comparisons of checkpoints or conditions, our results support four reporting practices. State the matching rule and show the headline contrast under at least one alternative rule. Fix and report the text window, and check whether generations reach the length cap. Include a control condition that shares the surface property most likely to trigger the rule, such as emotional content without a request. When relating markers to preference data, report the association with and without length terms, since a raw phrase count mixes a length component with a phrase component.

6 Conclusion

On identical OLMo-2 responses, the matching rule and the text window decide whether an empathy marker looks more or less common in personal advice than in an emotional control. Targeted controls separate markers that all look advice-specific against factual prompts, DPO moves the validation and structuring gaps in opposite directions, and length accounts for about half of the chosen-side advantage of referral and structuring phrases in the DPO data. These results describe detector outputs. Human labels and stored responses at every checkpoint would let the same design support claims about semantic behavior.

7 Limitations

All analyses are observational. Checkpoint and corpus comparisons do not attribute outputs to training

examples. Base uses plain QA while later checkpoints use chat serialization, confounding every Base-to-SFT contrast.

The 120-sentence audit uses LLM-consensus labels and has no independent human labels. It does not validate response-level prevalence, samples 97 seed-0 responses, and has wide category intervals. The detector-component and window analyses cover only final 7B, because raw responses for other checkpoints were not stored.

The targeted controls are not topic-paired to the advice prompts, and a non-significant contrast does not show equivalence. Twelve pilot prompts were reused; excluding them changes final-7B advice-minus-general-facts gaps by at most .045. The small seed sets do not remove sampling uncertainty.

The DPO length bins are post hoc. The conditional model was specified before its fits but after the aggregate results were known, so the analysis is exploratory and not preregistered. Conditioning on response length changes the estimand, and an OR is not a probability change. The mixtures share 21,508 content-identical pairs. SFT recoverability uses only three to six distinct three-word types. Stage-2 Dolmino-Mix-1124 lacks a public index, and Stage-1 novelty is verbatim-only and length-confounded. One model family, two sizes, one decoding configuration, and modest author-written prompt sets support no scaling or cross-family claim.

Ethics Statement

Data and human subjects. We use public corpora and open-weight OLMo-2 checkpoints. We did not recruit participants or collect new human annotations for the analyses reported here. The 140 prompts are author-written hypothetical situations without personal information.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns surface phrases that can mark safety-relevant moves such as professional referrals. Within the LLM-labeled audit sample, the detector produces many false positives; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety-relevant marker inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and provide an analysis tool, not any model or dataset that could cause direct harm.

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A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-p 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt in the canonical grid. The battery: 60 advice, 30 factual, 20 emotion-without-request prompts, 20 neutral how-to prompts, and 10 health-facts prompts; the complete machine-readable list ships as `experiments/prompts_full.json`. The Base/SFT/DPO five-condition extension uses four seeds; all balanced extension tests use seeds 0–3 for every condition. Eight advice and four general-facts prompts were reused from the earlier 12-prompt pilot; the other 128 are new to the full battery.

Examples.

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`
- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`
- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

B Surface-marker lexicon and detector threshold

This section records the four-category detector implemented in `experiments/raw_output_robustness.py`. An earlier variant also included a crisis category, which is excluded after its audit result. A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), with $\tau = 0.82$ primary. Every sentence is eligible; despite its name, `empathy_opener` is not restricted to an opening span.

Lexicon (substring match).

- **empathy-opener proxy** (`empathy_opener`): `i'm sorry to hear`, `i am sorry to hear`, `i'm really sorry`, `that sounds really`, `that sounds very`, `i can understand`, `i can imagine`, `it's understandable`, `that must be`, `i'm here for you`, `i hear you`, `it sounds like you`
- **validation-phrase proxy**: `your feelings are valid`, `it's okay to feel`, `it is okay to feel`, `it's normal to feel`, `it's completely normal`, `what you're feeling`, `there's nothing wrong with`, `it's natural to`
- **referral/disclaimer proxy** (`disclaimer`): `consult a`, `talk to a`, `speak with a`, `see a doctor`, `seek medical`, `i am not a doctor`, `i'm not a doctor`, `i'm not a medical`, `medical professional`, `healthcare professional`, `mental health professional`, `seek professional`, `professional help`
- **structuring-marker proxy** (`structure`): `here are some`, `here are a few`, `consider the following`, `steps you can take`, `a few things you can`, `first,`, `secondly,`

e5 seed exemplars. Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard"; "I can only imagine how painful this

is." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way"; "There's nothing wrong with feeling upset." Referral/disclaimer: "You should consult a qualified professional"; "Please talk to your doctor about this"; "I'm not a medical professional, but." Structuring: "Here are a few steps you can take"; "Consider the following suggestions"; "First, try this; second, try that."

C Threshold sweep

Table 8 reports the exact contrasts at $\tau = 0.80$, 0.82, and 0.84. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy's emotional-minus-advice point estimate changes sign at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was fixed before the 140-prompt battery, but its qualitative pattern should not be generalized across detector thresholds.

Model	Contrast	.80	.82	.84
1B	empathy: emot.−adv.	.030	.200	−.107
7B		.030	.143	−.280
1B	validation: adv.−max ctrl.	.030	.243	.107
7B		.047	.253	.170
1B	referral: adv.−domain	−.003	.080	.093
7B		.017	.093	.280
1B	structuring: adv.−neutral	.000	.000	.047
7B		.000	−.030	−.010

Table 8: Threshold robustness of the four targeted-control contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on personal advice and neutral how-to.

D Final-7B response-level robustness

The file `final_7b_answers.jsonl` contains all 700 final-7B responses (140 prompts × five seeds). Table 9 shows the opportunity imbalance. "Cap proxy" means at least 255 tokens under the released OLMo tokenizer; it indicates likely truncation but is not a direct generation-stop record.

Table 2 in the main text recomputes each targeted-control contrast using exact substring matching alone and the exact-or-embedding union, over all stored text and fixed prefix windows. Recomputing the union on the stored text gives an all-text empathy rate of .6133 on advice, compared with .617 in the detector outputs saved at generation time (Table 4). The difference is one positive response among 300; we report each source where it is used. The control rate is .7600. The gap's point estimate changes sign at 64 tokens. The all-text and 128-token contrasts remain uncertain, so these are not two established opposite effects. Exact-only matching instead makes structuring look advice-specific relative to neutral how-to; the embedding component removes that contrast.

Condition	n	mean tok.	mean sent.	cap proxy
advice	300	243.8	13.5	.880
factual	150	187.4	7.8	.573
emotional	100	253.5	10.9	.980
neutral how-to	100	254.6	13.8	.970
health facts	50	250.5	10.2	.960

Table 9: Final-7B response length and generation-cap proxy.

Among advice responses, union detections include one category in 4.7% of cases, two in 29.0%, three in 40.7%, and four in 25.7%. Pairwise phi correlations range from −.14 to .21; referral/disclaimer appears in 93.0% of validation-positive responses. Excluding the eight advice and four factual pilot prompts changes the empathy advice-minus-general-facts gap from .587 to .542, validation from .367 to .369, referral/disclaimer from .850 to .831, and structuring from .423 to .458.

E Stagewise rates

Table 10 supplies the rates underlying the transition results in Section 4.2. Parentheses contain the advice-minus-general- facts gap. The rates are detector outputs, not human judgments.

F Checkpoint interactions with targeted controls

Table 11 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted q in parentheses. The family contains all 24 displayed transition tests.

G Per-phrase recoverability and leave-one-out

Corpus searches provide secondary context. For each stage's available indexed data we search the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`;

Model	Stage	Empathy	Validation	Referral/disclaimer	Structuring
1B	Base	.217 (.204)	.073 (.073)	.483 (.430)	.757 (.430)
1B	SFT	.393 (.393)	.167 (.167)	.697 (.684)	.847 (.767)
1B	DPO	.550 (.510)	.380 (.380)	.933 (.873)	1.000 (.400)
1B	RLVR	.567 (.507)	.410 (.410)	.923 (.836)	1.000 (.440)
1B	Instruct	.590 (.550)	.363 (.356)	.900 (.820)	1.000 (.420)
7B	Base	.230 (.230)	.087 (.080)	.577 (.524)	.853 (.486)
7B	SFT	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	DPO	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	Instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 10: Personal-advice detector-positive rate, with the advice-minus-general- facts gap in parentheses ($\tau = 0.82$). Base uses plain QA; later checkpoints use chat serialization.

Model	Marker	Control	Base	SFT	DPO	Base→SFT	SFT→DPO
1B	referral	health facts	.188	.308	.167	+.121 (.414)	−.142 (.296)
1B	empathy	emotional	.017	.225	−.242	+.208 (.095)	−.467 (.001)
1B	structuring	neutral how-to	−.013	−.050	.000	−.037 (.653)	+.050 (.475)
1B	validation	health facts	.071	.146	.375	+.075 (.475)	+.229 (.095)
1B	validation	emotional	.021	.121	.225	+.100 (.293)	+.104 (.296)
1B	validation	neutral how-to	.058	.171	.350	+.113 (.189)	+.179 (.059)
7B	referral	health facts	.321	.067	.200	−.254 (.166)	+.133 (.293)
7B	empathy	emotional	.108	.467	−.175	+.358 (.016)	−.642 (.001)
7B	structuring	neutral how-to	.050	−.158	−.025	−.208 (.026)	+.133 (.095)
7B	validation	health facts	.092	.175	.317	+.083 (.475)	+.142 (.296)
7B	validation	emotional	.054	.125	.179	+.071 (.414)	+.054 (.537)
7B	validation	neutral how-to	.092	.163	.292	+.071 (.414)	+.129 (.189)

Table 11: Complete targeted-control checkpoint interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

novelty = 1−verbatim coverage), each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture-0225` for 1B; `tulu-3-sft-olmo-2-mixture` for 7B) in DuckDB with assistant-only scoping (866,138 conversations for 1B and 939,344 for 7B; oasst1 (Köpf et al., 2023) is not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares seeded marker phrases against length-matched topical controls from the same answers, per-million documents, with a distinct-phrase leave-one-out. Stage-2 mid-training uses Dolmino-Mix-1124, for which no public index was available; the corpus trace is therefore incomplete.

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	consult a	891	978	1.10
1B	it's understandable	218	227	1.04
1B	it's natural to	152	157	1.03
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 12: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types underlie the estimates in Appendix G; "here are some" dominates the 3-word mean.

Length-matched buckets (occurrence-weighted). Seeded-marker vs topical per-million for 1B: 2-

word 1884.3 vs 1760.2 (1.1×, n_{occ}=14), 3-word 12900.2 vs 629.8 (20.5×, 45), 4-word 1596.9 vs 11.7 (136.9×, 30). For 7B, 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

H DPO phrase and source robustness

Table 13 gives paired risk-difference and matched-odds-ratio intervals. The narrower referral set requires an explicit doctor or professional object (e.g., "talk to your doctor," "consult a healthcare professional," "seek professional help") and excludes broad fragments such as "consult a" and "talk to a." Its semantic precision has not been human-validated. Intervals use the discordant-pair normal approximation for the risk difference and log-MOR; the accompanying scripts also report exact McNemar p-values.

Table 14 stratifies pairs by their length difference. The near-antisymmetric signs in the longer-response strata show that the aggregate chosen-side effect is strongly associated with relative response length. The comparable-length band is descriptive and post hoc; it is not a substitute for the conditional model in Table 7.

In the aggregate analysis, broad-referral differences are positive in eight of nine 1B sources (one tie) and all nine 7B sources; narrower-referral differences are positive in six and seven sources. After flexible length conditioning, adjusted ORs exceed one in eight of nine sources for broad referral and structuring at both sizes, and six of nine for the narrower set. Their ranges are broad referral .919–1.294 (1B) and .988–1.584 (7B), narrower referral .600–1.537 and .636–2.242, and structuring .718–1.213 and .718–1.324. Source intervals can be wide, so these are heterogeneity descriptions rather than source-level discoveries. The 19,890 7B rows with missing source labels are retained as the null-source stratum.

There are 367,082 distinct 1B prompt texts and 367,110 distinct 7B prompt texts; 21,551 1B rows and 21,574 7B rows belong to duplicated groups, with maximum multiplicity 15. Prompt-cluster covariance and one-pair-per-prompt estimates address this exact-text dependence. A separate content hash finds 21,508 distinct pairs shared across mixtures (21,515 multiplicity-matched rows), limiting independent-replication language.

I Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign Dirichlet(1) weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 15 reports percentile intervals. These intervals quantify sampling uncertainty around the LLM-consensus labels.

J Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$. With $B = 10{,}000$, zero exceedances yield the minimum $p = 1/10{,}001 \approx 9.999 \times 10^{-5}$, not zero. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean seeded-marker per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs, Haldane–Anscombe matched odds ratios with log-scale intervals, paired risk differences with normal-approximation intervals, and phrase occurrences per million whitespace-delimited response words. The exploratory conditional fits use the no-intercept paired likelihood in Section 3.6, a five-knot restricted cubic spline of each side's log word count, and prompt-cluster sandwich covariance multiplied by $G/(G-1)\,(n-1)/(n-k)$. All aggregate and length-band counts were reproduced before fitting. A separate natural-spline implementation reproduced all six marker coefficients within 7.6×10^{-7} and their standard errors within 3.2×10^{-9}. The accompanying scripts derive all statistics from the artifact.

Files. Stagewise, condition, and corpus numbers derive from `summary_v3.json`; audit numbers, both annotation passes, consensus labels, and adjudication reasons are in `ai_consensus_validation_v3.json`. Paired DPO features and conditional estimates are under `output/revision/dpo_conditional/`. The detector-component contrasts are produced by `experiments/window_component_reanalysis_20261109.py`, with estimates and input hashes in `output/review_2026-11-09/component_reanalysis/paired_measurement_`

Mixture	Phrase group	Δ pp [95% CI]	MOR [95% CI]	Chosen only	Rejected only
1B	referral/disclaimer	.100 [.065,.135]	1.181 [1.114,1.252]	2478	2098
1B	narrower referral	.023 [.006,.040]	1.170 [1.039,1.318]	591	505
1B	structuring	1.393 [1.278,1.507]	1.242 [1.220,1.264]	27047	21779
7B	referral/disclaimer	.179 [.144,.214]	1.352 [1.274,1.434]	2602	1925
7B	narrower referral	.043 [.026,.061]	1.335 [1.187,1.502]	649	486
7B	structuring	2.280 [2.167,2.394]	1.435 [1.409,1.461]	28466	19838

Table 13: DPO paired absolute and relative effects. Positive Δ means more chosen-only than rejected-only matches.

Mixture	Word-difference band	n	Referral		Narrower referral		Structuring	
			Δ pp	MOR	Δ pp	MOR	Δ pp	MOR
1B	chosen longer > 20	171170	+.470	1.991	+.059	1.408	+5.979	2.361
1B	rejected longer > 20	134707	−.310	.616	−.020	.871	−3.786	.579
1B	within ± 20	72424	−.010	.965	+.017	1.238	+.184	1.062
7B	chosen longer > 20	187201	+.502	2.172	+.098	1.761	+6.562	2.610
7B	rejected longer > 20	120381	−.223	.706	−.015	.905	−3.239	.619
7B	within ± 20	70759	+.009	1.028	−.003	.966	+.343	1.123

Table 14: DPO effects stratified by chosen-minus-rejected response length. Positive Δ means more chosen-only matches.

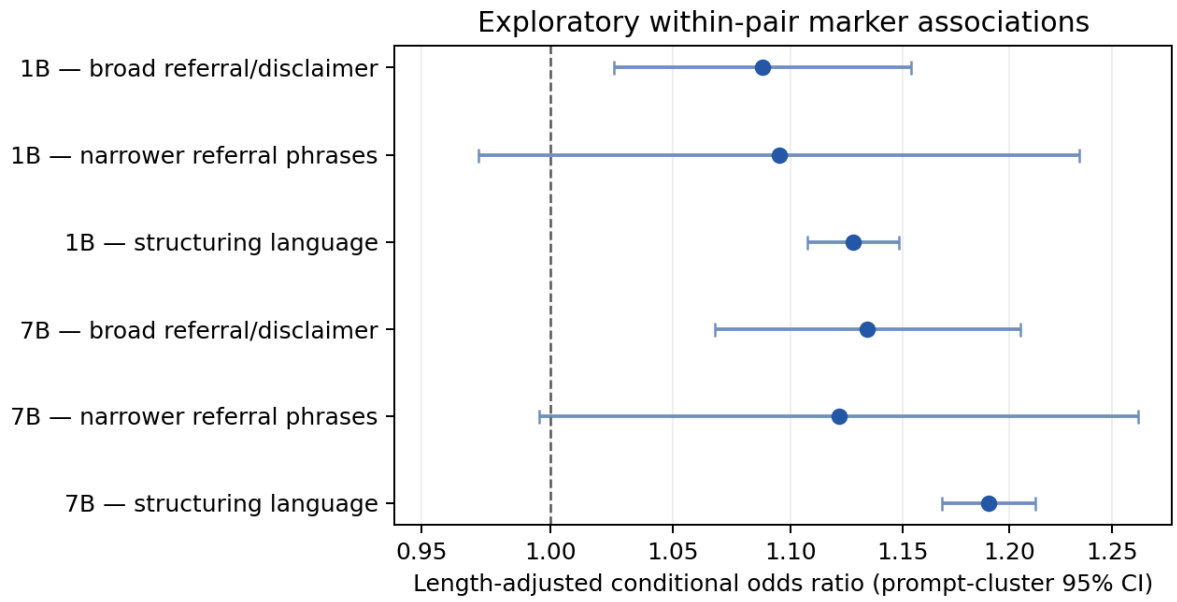


Figure 2: Paired conditional marker associations in the 1B/7B DPO training mixtures after flexible response-length modeling. Bars are 95% prompt-cluster sandwich intervals. OR $= 1$ denotes no modeled marker–choice association, not a zero change in preference probability.

`contrasts.json`. Figure 1 is produced by `experiments/make_main_figure.py`.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]

Table 15: Bayesian response-cluster bootstrap intervals for the 120-sentence LLM-consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.